INCH-POUND

MIL-T-47326(MI)
5 January 1990

MILITARY SPECIFICATION

TAPE, PRESSURE-SENSITIVE ADHESIVE, ALUMINUM-BACKED, ELECTRICALLY CONDUCTIVE

This specification is approved for use by the Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of aluminum foil electrical tape with an electrically conductive pressure-sensitive adhesive.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

FEDERAL

PPP-T-680	Tape, Pressure-Sensitive Adhesive, Packaging and Packing of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5999

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STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-45662	Calibration Systems Requirements

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D1000	Pressure-Sensitive Adhesive Coated Tapes Used for Electrical Insulation

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

(Non-Government standards and other publications are normally available from the organization which prepares or which distributes the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of the document and the references cited herein the text of this document takes precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.2), in accordance with 4.4.

3.2 Materials. The material shall be an electrically conductive, pressure-sensitive, aluminum-backed adhesive tape (see 4.6.2).

3.3 Roll core. The tape shall be furnished on cores made of paper fiber or non-fibrous plastic. The core shall have sufficient rigidity to prevent distortion of the roll under normal conditions of transportation, storage, and use. Unless otherwise specified, the inside diameter of the core shall be 1.500 (± .062) inches, and the width shall correspond to the width of the tape. The inside area of the core shall be of sufficiently light color to support identification marking. When specified in the contract or purchase order, a non-adhering liner shall be placed between layers of the tape (see 4.6.3 and 6.2).

3.4 Width and length. The width shall be as indicated in 6.4. Length shall be 18 (± 0.125) yards (see 4.6.4).

3.5 Film thickness. (See 4.6.5).

3.5.1 Overall thickness. The overall thickness of the tape shall be 0.0045 ± 0.001 inch.

3.5.2 Backing thickness. The aluminum foil backing thickness of the tape shall be 0.0020 ± 0.0005 inch.

3.6 Adhesion to steel. The tape shall provide adhesion to steel strength of 25 ounces/inch width, minimum (see 4.6.6).

3.7 Tensile strength. The tape shall display a tensile strength of 10 pounds/inch width, minimum (see 4.6.7).

3.8 Electrical resistance. The electrical resistance of the tape shall be 0.01 ohms/inch^2, maximum (see 4.6.8).

3.9 Fungus. Unless otherwise specified (see 6.2), the fungus conditions shall consist of exposure of one roll of tape and one 12-inch loose strip of tape with liner removed to the fungus cultures shown in table I for 28 continuous days at an ambient temperature of 30°(86°F) and a relative humidity of 95 ± 5 percent. During and after exposure, the tape shall show no evidence of supporting fungus growth (see 4.6.9).

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TABLE I. Fungus cultures.

Organism	American type culture Collection Number 1/	Quartermaster Number 2/
Chaetomium globosum	6250	359
Aspergillus niger	6275	458
Aspergillus flavus	10836	1223
Penicillium citrinum	9849	1226

1/ Source: American Type Culture Collection
2112 M Street, N.W.
Washington 6, D.C.

2/ Source: Mycology Laboratory, PRD, Quartermaster
Research and Engineering Center
Natick, Massachusetts

3.10 Storage life. The tape, in the original, unopened, individual container, shall meet the requirements of 3.2.3, 3.2.4, and 3.3 after storage for up to one year at temperatures in the range of 25° + 10°C (77° + 18°F) and with relative humidity not exceeding 90 percent (see 4.6.10).

3.11 Marking identification. Marking for identification shall be in accordance with MIL-STD-130, shall appear on the inside circumference of the roll core, and shall include, but not be limited to, the following (see 4.6.1).

a. The part or identifying number (PIN).

b. The manufacturer's name or trademark and CAGE code.

c. Date of manufacturer.

3.12 Workmanship. The tape shall be clean and free from foreign matter and defects that may impair its serviceability or appearance. The tape shall be smooth, uniform, and free from lumps or bare spots. The tape edges shall be straight and unbroken (see 4.6.1).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the following test conditions.

Temperature:	70 to 80°F
Humidity:	75 percent maximum
Barometric pressure:	Local corrected to sea level.

4.4 First article inspection.

4.4.1 First article sample. When specified in the contract or purchase order (see 3.1 and 6.2), the first article sample shall consist of five rolls of tape representative of the production product. The first article sample shall be subjected to the inspections and tests as specified herein in the order shown. Quantity production shall be withheld until the first article sample has been pronounced satisfactory by the procuring activity.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the inspections listed in table II and specified herein.

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TABLE II. Quality conformance inspection.

Inspection	Requirement Paragraph	Test Paragraph
Material	3.2	4.6.2
Roll core	3.3	4.6.3
Width and length	3.4	4.6.4
Thickness	3.5	4.6.5
Adhesion to steel	3.6	4.6.6
Tensile strength	3.7	4.6.7
Electrical resistance	3.8	4.6.8
Fungus	3.9	4.6.9
Storage life	3.10	4.6.10
Marking for identification	3.11	4.6.1
Workmanship	3.12	4.6.1
Packaging and packing	5.1	4.6.1
Marking	5.2	4.6.1

4.5.1 Lot size. Lot size shall consist of all the tape manufactured in one production batch run, from one process, forming part of one contract or order, and submitted for acceptance at one time and place.

4.5.2 Quality conformation sample. Quality conformance inspection shall be performed on 2 rolls of tape taken from each lot offered for acceptance.

4.5.2.1 Action in case of failure. If the sample fails to meet one or more of the specified requirements for quality conformance inspection, the entire lot represented by the failed sample shall be subjected to rejection. Subsequent lots shall not be offered for acceptance until technical judgment dictates that the cause of failure has been analyzed and solved to the satisfaction of the procuring activity.

4.5.3 Inspection equipment and facilities. Inspection equipment and facilities shall be of adequate quality and quantity to permit performance of the required inspections.

4.5.3.1 Inspection equipment accuracy. Equipment used to measure test conditions and test parameters shall be calibrated in accordance with MIL-STD-45662.

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4.5.3.2 Inspection conditions. Unless otherwise specified, all inspections shall be conducted under the following conditions of ambient temperature, humidity, and barometric pressure:

a. Temperature: 25 ± 10°C (77 ± 18°F)

b. Relative humidity: 90 percent, maximum

c. Barometric pressure: 30 ± 2 inches of mercury

4.6 Test methods.

4.6.1 Visual inspection. Visual inspection shall verify compliance with 3.11, 3.12, 5.1, and 5.2.

4.6.2 Material. Visually observe for texture and backing. Perform tests specified, in 4.6.6 and 4.6.8. Successful completion of these tests shall verify compliance with 3.2.

4.6.3 Roll core. Perform visual and tactile examination to determine rigidity and color. Using standard measuring equipment, measure width and inside diameter. Verify compliance with 3.3.

4.6.4 Width and length. A test specimen, not less than 18 inches in length, shall be removed from a sample roll and placed adhesive side up on a smooth, flat surface. The test specimen shall be permitted to relax to remove any latent elongation induced by the tension exerted on the tape during removal from the roll. The width of the relaxed specimen shall then be measured to the nearest 1/64-inch, using a standard steel scale. Ten measurements shall be made, uniformly distributed along the length of the specimen. The width of the sample shall be the average of ten measurements made on the specimen. Randomly select a roll of tape and measure length, using standard measuring equipment. Verify compliance with 3.4.

4.6.5 Thickness. The thickness of the tape shall be determined in accordance with ASTM D1000. Verify compliance with 3.5.

4.6.6 Adhesion to steel. The test for adhesion to steel shall be in accordance with ASTM D1000. Verify compliance with 3.6.

4.6.7 Tensile strength. The tensile strength of the tape shall be determined in accordance with ASTM D1000. Verify compliance with 3.7.

4.6.8 Electrical resistance. Resistance measurements shall be performed in accordance with MIL-STD-202, method 303. The electrical resistance through the tape shall be determined using an apparatus as shown in figure 1. The brass electrodes shall be one inch square and 0.125 (minimum) inches thick, and shall

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be smooth and free of burrs and scratches. After solvent cleaning of the electrodes, the tape shall be applied to one of them using contact pressure only. The power supply shall be adjusted to deliver 0.10 ± 0.01 ampere of current and applied for 5 ± 1 seconds. Place the electrodes together as shown and apply a 5.0 ± 0.5 pound load. Measure the voltage drop between the electrodes as soon as possible after the load has been applied. Calculate the electrical resistance. Tape specimens shall be not less than one inch wide. Verify compliance with 3.8.

$$\text{Electrical resistance (ohms/in.}^2\text{)} = \frac{\text{voltage drop}}{\text{0.1 ampere}}$$

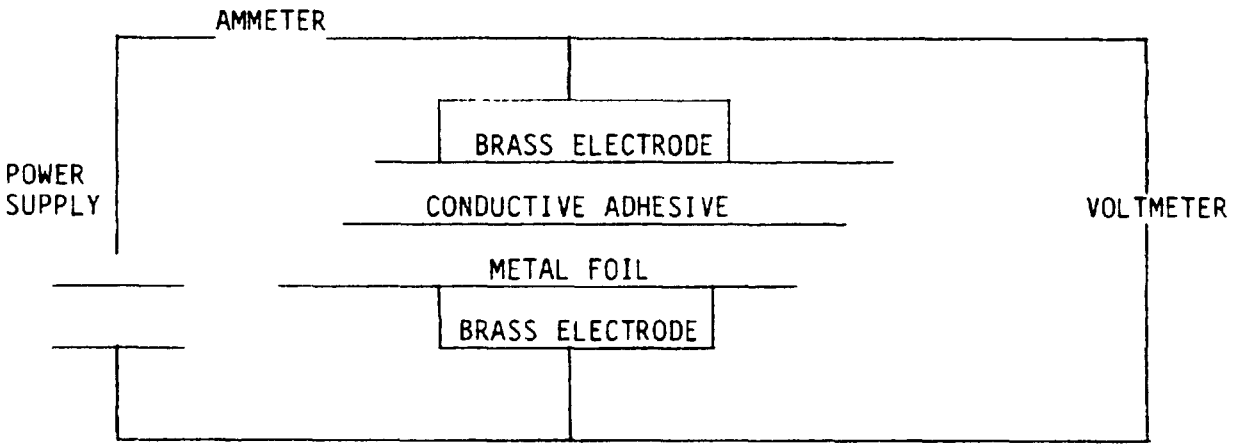


Figure 1 - Electrical Resistance Schematic Diagram

4.6.9 Fungus. Unless otherwise specified (see 6.2), the tape shall be tested in accordance with MIL-STD-810, method 508, procedure I or II, except that the conditions of 3.9 shall apply. After completion of the exposure, compliance with 3.9 shall be verified. The procuring activity, at its discretion, may accept a certification of the use of fungus resistant material in lieu of testing.

4.6.10 Storage life. The contractor shall furnish certification and, upon request, supporting test data verifying compliance with the requirements of 3.10.

5. PACKAGING

5.1 Packaging and packing. Unless otherwise specified, packaging and packing shall be in accordance with the requirements of PPP-T-680, at the level specified in the contract or purchase order (see 4.6.1).

5.2 Marking. Containers of the tape shall be marked for shipment and storage in accordance with the requirements of MIL-STD-129 (see 4.6.1).

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6. NOTES

6.1 Intended use. The tape covered by this specification is intended for use in applications where static dissipation of electrical continuity is desired.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation, and if required the specific issue of individual documents referenced (see 2.1)

c. PIN

d. Quantity (number of rolls)

e. Whether a first article sample is required (see 3.1 and 4.4)

f. Certification of fungus resistant materials in lieu of testing, if permissible (see 3.5 and 4.5)

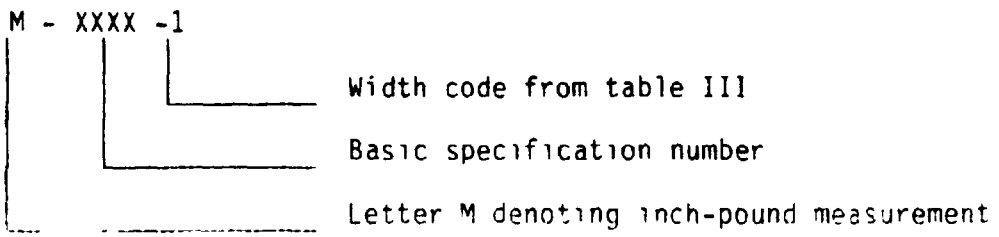
g. Certification of storage life capabilities (see 3.6 and 4.6)

h. Whether a liner is required

6.3 First article. When a first article inspection is required, the items should be a first article sample. The first article should consist of five rolls of tape. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Part or identifying number (PIN). The PIN shall consist of the letter M, plus the basic number of this specification, followed by a number indicating tape width (see table III).

Example:



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TABLE III. PIN codes.

Width in inches	Tolerance in inches	PIN Code
0.500	± 0.03	1
0.750	± 0.03	2
1.000	± 0.06	3
1.250	± 0.06	4
1.500	± 0.06	5
1.750	± 0.06	6
2.000	± 0.06	7

6.5 Metrication. Wherever inch-pound units of measurement appear in this specification, metric equivalents in accordance with FED-STD-376 shall be acceptable.

6.6 Subject term (keyword) listing.

Tape
Electrical tape

Custodian:
Army - MI

Preparing activity:
Army - MI

(Project No. 5999-A101)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-47326	5 Jan 1990

3. DOCUMENT TITLE
Tape, Pressure-Sensitive Adhesive, Aluminum-Backed, Electrically Conductive

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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